



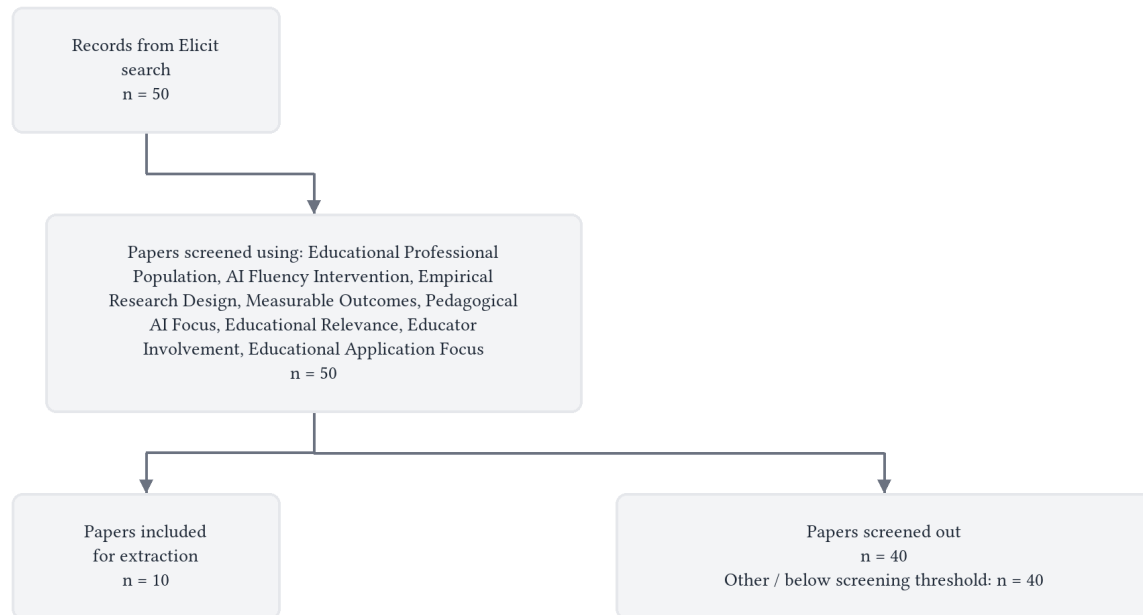


## Abstract

Multiple complementary frameworks for educator AI fluency converge on the core philosophy of AI as augmentation rather than automation, with human agency remaining central to pedagogical decision-making [1–3]. The explicit “4D” framework—encompassing Delegation, Description/Prompting, Discernment, and Diligence with progression levels of Acquire (literacy), Deepen (fluency), and Create (mastery)—appears in computing education contexts [4], while other disciplines employ structurally similar models emphasizing the same core competencies under different terminology [1, 2, 5]. Evidence confirms all three UNESCO competency domains in practice: educators use AI for differentiated instruction, project-based learning, and assessment design (pedagogy) [6]; for content creation, grading automation, and feedback loops (operations) [1, 2, 4]; and for bias detection, data privacy protection, and academic integrity maintenance (ethical fluency) [1, 2, 5]. The three-tier progression path appears consistently across contexts [1, 2, 4, 5], though implementation varies substantially by institutional capacity—from 75% faculty discomfort with minimal integration [7] to intensive multi-year professional development programs with 40+ training sessions [1].

The AI-fluent classroom checklist elements find empirical support: transparent use practices include stoplight frameworks guiding appropriate AI citation [1], critique sessions involve “fact-check the AI” exercises and building students’ ability to evaluate AI responses critically [8], and human-first assessment emphasizes process over product with invigilated demonstrations [4]. However, significant barriers persist, including technical infrastructure gaps [1, 7], inadequate professional development lacking practical guidance on societal implications [7], and ethical concerns about bias, data privacy, and academic integrity [2, 4, 7]. Successful implementations consistently embed ethical frameworks from inception rather than treating them as afterthoughts [1, 2, 5], balance productivity gains with guardrails preventing over-reliance [3, 8], and adapt shared competency models to disciplinary contexts while maintaining focus on the four core pillars of delegation, prompting, discernment, and diligence [3–5].

## Flow Diagram



## Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "AI Fluency for Educators

I. The Core Philosophy: AI as a Co-Pilot • The Shift: Moving away from automating compliance (grading, attendance) toward augmenting creativity and critical thinking. • The Golden Rule: Human agency remains central. AI provides the "draft," but the educator provides the "discernment." II. The "4D" Framework for AI Fluency This model (adapted from 2025-2026 educational standards) defines the four pillars of a fluent educator:

1. Delegation: Identifying which tasks are high-value for humans (mentorship, emotional support) vs. which are high-value for AI (differentiating reading levels, generating initial lesson outlines).
2. Description (Prompting): Moving beyond basic chat to Prompt Engineering. Using context, persona, and iterative feedback to get high-quality pedagogical outputs.
3. Discernment: The ability to audit AI output for "hallucinations," bias, or lack of cultural relevance. Fluent educators don't just "copy-paste"; they verify.
4. Diligence: Maintaining ethical standards, data privacy, and academic integrity. Knowing the "why" and "how" of AI safety. III. Competency Domains (Based on UNESCO 2024/25 Standards)
5. AI Pedagogy & Design • Differentiated Instruction: Using AI to instantly translate a single lesson plan into five different Lexile levels or languages. • Socratic Tutoring: Setting up AI "bots" (like Khanmigo) to act as

tutors that guide students rather than giving them answers. • Assessment 3.0: Moving from "product-based" grading (essays) to "process-based" grading (discussing the AI-human collaboration process).

6. AI Operations & Productivity • Workflow Automation: Using AI to summarize parent-teacher meetings, organize lesson resources in Notion, or generate slide decks in Canva. • Feedback Loops: Utilizing AI to provide immediate, formative feedback on student drafts before they reach the teacher's desk.
7. Ethical & Social Fluency • Bias Awareness: Recognizing when a model reflects Western-centric or gendered biases and using that as a "teachable moment." • Data Privacy: Understanding what data is safe to feed into a LLM (Large Language Model) and protecting student PII (Personally Identifiable Information). IV. The Progression Path: From "Acquire" to "Create" Level Goal Educator Action Acquire Literacy Understanding basic terms (LLM, Hallucination) and using AI for basic email/admin tasks. Deepen Fluency Integrating AI into specific lesson plans; using AI as a "thinking partner" for curriculum design. Create Mastery Designing custom AI-driven learning environments and leading school-wide ethical AI policies.

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V. The "AI-Fluent" Classroom Checklist • [ ] Transparent Use: Students are taught how to cite AI and discuss its role in their work. • [ ] Critique Sessions: Lessons include "Fact-check the AI" exercises. • [ ] Human-First Assessment: Assessments prioritize oral exams, in-class demonstrations, and creative synthesis that AI cannot replicate. VI. Summary Quote for Reflection "AI will not replace teachers, but teachers who are AI-fluent will replace those who are not." — Common 2026 Educational Sentiment. "

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

## Screening

We screened in sources based on their abstracts that met these criteria:

- **Educational Professional Population:** Does the study involve educational professionals (teachers, educators, instructional designers, or educational administrators) at any educational level (K-12, higher education, professional development)?
- **AI Fluency Intervention:** Does the study examine training programs, professional development, or educational initiatives focused on developing AI literacy, AI integration skills, or AI pedagogical competencies?
- **Empirical Research Design:** Does the study use an empirical research design (quantitative, qualitative, or mixed-methods including experimental designs, quasi-experimental studies, observational studies, case studies, systematic reviews, or meta-analyses)?
- **Measurable Outcomes:** Does the study include clearly defined outcome measures related to AI competency, educator effectiveness, or educational impact?
- **Pedagogical AI Focus:** Does the study focus on AI tools used for educational purposes (such as lesson planning, assessment, differentiated instruction, or feedback systems)?
- **Educational Relevance:** Is the study focused on educational applications rather than AI in healthcare, business, or other non-educational sectors without clear educational relevance?
- **Educator Involvement:** Does the study include educator involvement or professional development components (rather than examining only student AI use)?
- **Educational Application Focus:** Does the study focus on educational applications or pedagogy rather than solely on AI algorithm development or technical specifications without clear educational application?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 40

## Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **AI Fluency Framework:**

Extract how AI fluency for educators is defined, conceptualized, or operationalized in this study, including:

- Specific competency models or frameworks used (e.g., 4D framework: Delegation, Description/Prompting, Discernment, Diligence)
- Key dimensions or components of AI fluency identified
- Progression levels or stages (e.g., Acquire/Literacy → Deepen/Fluency → Create/Mastery)
- How the study distinguishes AI fluency from general digital literacy

- **Educational Context:**

Extract details about the educational setting and participants:

- Education level (K-12, higher education, teacher training)
- Subject area or discipline focus
- Geographic location and institutional context
- Participant characteristics (teacher experience, prior AI exposure, demographics)
- Class/program size and duration

- **AI Tools and Implementation:**

Extract specific details about AI technologies and their educational implementation:

- Specific AI tools, platforms, or models used (e.g., ChatGPT, custom GPTs, LLM-powered assistants)
- How AI is integrated into teaching workflow (lesson planning, assessment, student support)
- Technical implementation details (prompting strategies, customization, controls)
- Human-AI interaction patterns observed

- **Competency Development Approach:**

Extract how AI fluency competencies are developed or acquired:

- Training programs, professional development models, or learning interventions used
- Progression pathways or scaffolding approaches
- Time investment and intensity required
- Role of peer collaboration, mentoring, or community of practice
- Self-directed vs. formal learning approaches

- **Educator AI Practices:**

Extract specific examples of how educators use AI in their professional practice:

- Pedagogical applications (differentiated instruction, assessment design, feedback)
- Administrative/productivity uses (lesson planning, grading, communication)

- Student-facing AI integration approaches
- Evidence of the 4D competencies in action (delegation decisions, prompting quality, discernment practices, ethical diligence)

- **Barriers and Facilitators:**

Extract factors that help or hinder AI fluency development for educators:

- Technical barriers (infrastructure, access, training gaps)
- Institutional factors (leadership support, policy, resources)
- Individual factors (mindset, self-efficacy, resistance to change)
- Ethical and safety concerns that impact adoption
- Successful implementation strategies and enablers identified

- **Impact on Teaching Practice:**

Extract evidence of how AI fluency affects educators' teaching effectiveness and practice:

- Changes in instructional design or delivery methods
- Impact on teacher workload, efficiency, or job satisfaction
- Effects on teacher-student relationships and classroom dynamics
- Evidence of enhanced pedagogical capabilities
- Unintended consequences or challenges that emerged

- **Student Learning Outcomes:**

Extract evidence about how educator AI fluency impacts student learning and experiences:

- Academic achievement or learning outcome measures
- Student engagement, motivation, or satisfaction changes
- Development of student AI literacy or critical thinking skills
- Effects on personalization, accessibility, or equity in learning
- Student perspectives on AI-enhanced instruction

- **Ethical AI Framework:**

Extract how ethical AI use and safety considerations are addressed:

- Specific ethical guidelines, policies, or frameworks applied
- Data privacy and student PII protection measures
- Bias detection and mitigation strategies implemented
- Transparency and explainability practices with students
- Academic integrity policies and enforcement approaches
- Professional responsibility and accountability structures

## Results

### Characteristics of Included Studies

| Study                                    | Full text retrieved? | Education Level                        | Subject Area/Discipline                       | Geographic Context                                           | Sample/Participants                                                                                                          |
|------------------------------------------|----------------------|----------------------------------------|-----------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| M. Neumann & Cynthia Gerstl-Pepin, 2025  | No                   | Higher education [7]                   | Human services and education [7]              | Large public university in the US [7]                        | 24 faculty members surveyed, 6 interviews, 3 professional development sessions observed [7]                                  |
| T. Chiu et al., 2024                     | Yes                  | K-12 [5]                               | AI education (AI literacy and competency) [5] | Hong Kong middle schools [5]                                 | 30 experienced AI teachers, average age 32, minimum 3 years teaching AI [5]                                                  |
| Michelle J. Kelley & Taylor Wenzel, 2025 | Yes                  | Higher education, teacher training [1] | Teacher education, reading education [1]      | University setting, likely University of Central Florida [1] | Graduate and undergraduate students in reading practicum (n=49 surveyed) [1]                                                 |
| J. Prather et al., 2023                  | Yes                  | Higher education [4]                   | Computing education [4]                       | Global, across 20 countries and five continents [4]          | Computing students and instructors with varying LLM experience [4]                                                           |
| Alex X. Liu et al., 2025                 | Yes                  | K-12, teacher preparation programs [6] | Literacy, STEM, social studies [6]            | Not explicitly mentioned [6]                                 | K-12 teachers, school leaders, paraprofessionals; 140,000+ teacher-AI interactions analyzed [6]                              |
| Miroslava Nadkova Petrova, 2026          | Yes                  | Higher education [2]                   | AI integration across disciplines [2]         | Global perspective [2]                                       | Diverse backgrounds in educational technology, curriculum design, ethics, leadership; mid-to senior-career professionals [2] |
| V. Karaban & A. Karaban, 2025            | Yes                  | Higher education [3]                   | Translation education [3]                     | Not mentioned [3]                                            | Not mentioned [3]                                                                                                            |



| Study                             | Full text retrieved? | Education Level                                    | Subject Area/Discipline       | Geographic Context                          | Sample/Participants                                                                          |
|-----------------------------------|----------------------|----------------------------------------------------|-------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------|
| Majeed Kazemitabaar et al., 2024  | Yes                  | Higher education [8]                               | C and Systems Programming [8] | Large North American university [8]         | 700 university students (56% male, 30% female); 8 programming educators from 6 countries [8] |
| Anastassia Zabrodska et al., 2026 | Yes                  | Higher education [9]                               | Various disciplines [9]       | Global, universities on five continents [9] | Not mentioned [9]                                                                            |
| Ngongpah, Guilen & O. Oni, 2025   | No                   | Teacher training and professional development [10] | Not mentioned [10]            | Not mentioned [10]                          | Not mentioned [10]                                                                           |

The included studies span diverse educational contexts, from K-12 to higher education, with particularly strong representation from teacher education and professional development programs. Geographically, studies represent Asia, North America, and Europe, with two global multi-country analyses [4, 9]. Sample sizes varied substantially, from focused qualitative studies with 24-30 participants [5, 7] to large-scale analyses involving 700 students [8] or 140,000+ educator-AI interactions [6]. Eight studies provided full-text access, while two were available as abstracts only [7, 10].

## AI Fluency Frameworks and Competency Models

Studies employed varying conceptualizations of AI fluency, though several converged on similar structural elements. The Digital Education Council's AI Literacy Framework emerged as a notable model, defining five dimensions with specific progression levels: Level 1 (foundational applied AI awareness), Level 2 (AI application in teaching and learning), and Level 3 (strategic AI leadership in higher education) [1]. This framework emphasizes strategic integration of AI into educational practices with attention to ethical use and leadership [1].

One study explicitly referenced a "4D" framework encompassing Delegation, Description/Prompting, Discernment, and Diligence, with progression levels of Acquire (literacy), Deepen (fluency), and Create (mastery) [4]. This framework distinguishes AI fluency from general digital literacy by focusing on AI-specific competencies like prompt engineering, discernment of AI output, and ethical considerations [4].

UNESCO's AI competency frameworks for students and educators provided another structural model, organized around human-centered mindset, ethics, AI techniques, and system design, with the same three-tier progression (acquire, deepen, create) [2]. This framework emphasizes universal AI literacy including prompting, bias detection, and ethics [2].

For K-12 contexts, a comprehensive framework proposed five key components: technology, impact, ethics, collaboration, and self-reflection [5]. This framework distinguishes AI competency from literacy by incorporating confidence and self-reflective mindsets alongside knowledge [5]. The framework identifies five effective learning experiences

to foster these abilities: community engagement, case studies, hands-on activities, exhibitions, and cultural learning [5].

Translation education offered a domain-specific model using custom GPTs structured around pre-production, production, and post-production phases of professional workflows, with emphasis on AI literacy and prompt engineering as essential educator competencies [3]. This approach operates within a human-centered AI paradigm where technology augments rather than replaces human capabilities [3].

Several studies did not employ explicit formal frameworks but addressed AI fluency through emergent themes related to pedagogical practices, critical perspectives, and ethical considerations [9].

## Competency Development Approaches

Professional development models varied considerably across contexts. In higher education settings, faculty attended professional development trainings that addressed GenAI functionality and raised ethical concerns, though notably failed to provide practical guidance on discussing societal implications with students [7]. One teacher education program documented intensive engagement over two years, with faculty attending over 40 AI training sessions [1]. This institution formed a Special Interest Group and hosted an AI institute for K-12 practitioners, demonstrating a peer collaboration approach [1].

K-12 contexts employed iterative co-design approaches, with 30 experienced AI teachers participating in workshops and multiple cycles of data collection and analysis [5]. The emphasis on community engagement, hands-on activities, and presentations created pathways for continuous learning [5].

For computing education, a shift toward using LLMs as partners in pair programming emerged, with the tools themselves providing scaffolding and detailed explanations that support self-directed learning [4]. This represents a pedagogical pivot from traditional methods to using LLMs for problem-solving while generating personalized teaching materials to reduce instructor workload [4].

A structured roadmap for higher education included embedded first-year courses covering prompt engineering, bias detection, and ethics, coupled with discipline-specific integration and microcredentialing [2]. The model proposed an "AI Wrangler" certification program to support faculty adoption [2]. This approach combined mandatory first-year courses with discipline-specific modules, representing formal learning structures [2].

Translation programs implemented structured integration of custom GPTs into curricula, fostering AI literacy and prompt engineering skills through role-based assistants that enhance strategic thinking and decision-making [3]. This approach balanced front-loaded design work with back-loaded savings in instructor workload, combining structured guidance with autonomous practice [3].

Programming education deployed AI assistants with semester-long implementations featuring iterative design and frequent feedback loops [8]. Features like pseudo-code generation and code annotation created progression pathways from basic usage to more complex tasks, supporting self-directed learning through interactive features [8].

Global recommendations emphasized robust professional development that must be continuous and contextualized, with school leadership cultivating cultures supporting innovation [10]. The integration of AI literacy into teacher education programs represented formal learning pathways [10].

## **Educator AI Practices Across Pedagogical Domains**

### **Pedagogical Applications**

Educators demonstrated diverse AI applications for instructional enhancement. In K-12 settings, teachers used AI for differentiated instruction, project-based learning, critical thinking development, explicit teaching, and establishing instructional routines [6]. The framework's five components—technology, impact, ethics, collaboration, and self-reflection—guided integration approaches emphasizing confidence and self-reflective mindsets [5].

Teacher education programs implemented AI for personalizing learning and providing tutoring support, with specific applications in source evaluation, literature review, and data table creation [1]. A "stoplight framework" guided appropriate use: green indicating permissible use with citation, yellow for grammar and mechanics only [1]. This transparent approach promoted ethical practices while supporting pedagogical aims [1].

Computing educators used LLMs for code generation and interpretation, pair programming, and providing detailed code explanations [4]. The emphasis shifted toward process over product, with invigilated assessments representing new student-facing integration approaches [4].

Higher education broadly employed AI for customized educational content, innovative teaching methods, and technology-enhanced assessment, with intelligent tutoring systems and chatbots providing real-time feedback and personalized learning [2]. Educators developed universal AI literacy through structured interactions fostering critical thinking [2].

Translation education structured custom GPTs into curricula to enhance strategic thinking and decision-making, with role-based GPTs supporting lesson planning and curriculum design [3]. The teacher-machine-student triad encouraged self-directed practice while maintaining human oversight and ethical considerations [3].

### **Administrative and Productivity Uses**

AI enhanced instructional practices, content creation and adaptation, and assessment and feedback loops [6]. Teachers automated tasks like assessment, plagiarism detection, and feedback administration, allowing focus on strategy and experimentation [2].

Computing educators generated teaching materials and assignments using AI [4], while teacher educators streamlined grading and feedback processes and used AI for planning and adaptation [1]. The emphasis on AI for these productivity gains appeared consistent across contexts.

### **Evidence of 4D Competencies in Practice**

Delegation decisions manifested in educators' strategic choices about task allocation. Teachers used AI for tasks like summarizing sources while maintaining control over higher-order decisions [1]. Programming educators considered throttling usage to prevent misuse, demonstrating careful delegation boundaries [8].

Prompting quality emerged as a critical competency, with educators framing pedagogically grounded prompts and developing specific skills in prompt design [6]. The pseudo-code feature in programming tools was appreciated for structured learning experiences [8].

Discernment practices included evaluating AI output for hallucinations and biases [1]. Programming educators emphasized building students' ability to critique AI responses critically [8]. Teachers demonstrated emerging competencies in instructional adaptation and critical evaluation [6].

Diligence manifested through emphasis on ethical standards and data privacy [1]. Educators emphasized transparency in AI responses and ensuring academic integrity [8]. The integration of critical perspectives reflected discernment practices and ethical diligence across multiple contexts [9].

## **Barriers and Facilitators to AI Fluency Development**

### **Technical Barriers**

Limited understanding of GenAI functionality represented a fundamental technical barrier [7]. In K-12 contexts, current AI literacy definitions developed from engineering perspectives proved unsuitable for non-technical audiences [5]. Translation education faced limited AI integration in curricula, with risks of hallucinations and biases without careful prompt design and human oversight [3].

Programming education required "guardrails" to prevent direct solutions, highlighting infrastructure and access challenges [8]. The sensitivity of LLM output quality to prompt formatting and schema specificity created technical barriers at scale [6]. Technological infrastructure gaps in schools, particularly unequal access to AI resources, advanced hardware, utilities, and reliable internet connectivity, emerged as critical constraints [1].

### **Institutional Factors**

Professional development opportunities frequently lacked practical guidance on discussing societal implications [7]. K-12 frameworks had not been tested in field settings, with teacher capacity affecting learning design and development [5]. Computing education faced needs for clear policies on generative AI use, with potential to undermine academic integrity [4].

Successful institutional factors included establishment of university policies and guidelines [1], creation of Special Interest Groups for AI literacy [1], and leadership support through professional learning efforts [1]. However, rigid curricula and faculty resistance presented significant barriers [2], along with lack of training, resources, and limited AI expertise among teaching staff [2].

### **Individual Factors**

Faculty discomfort or neutrality toward GenAI stemmed from lack of understanding or ethical concerns [7]. Teacher perspectives proved crucial, with self-reflective mindsets important for lifelong learning [5]. Resistance to technological change appeared widespread, though positive mindsets and technology self-efficacy served as facilitators [10].

Computing education identified potential for over-reliance on LLMs, leading to reduced motivation and self-understanding [4]. Individual concerns about plagiarism, data privacy, and security represented resistance to change due to ethical concerns [1]. Faculty resistance and student overreliance on AI emerged as individual barriers requiring attention [2].

### **Ethical and Safety Concerns**

Potential for biased and misinformation, energy consumption, academic integrity violations, data privacy breaches, and intellectual property rights issues constituted major ethical concerns [7]. K-12 contexts required continuous evaluation of AI understanding with self-reflective mindsets critical for ethical use [5].

Computing education faced potential for biased or incorrect answers and introduction of security vulnerabilities [4]. Higher education broadly grappled with bias in algorithms, data privacy issues, and unequal access to resources [1].

Concerns about bias in AI outputs, data privacy, and socioeconomic disparities appeared across multiple contexts [2].

Translation education highlighted data privacy and potential biases as paramount ethical issues [3]. Programming education addressed trusting AI responses, potential misuse, and data privacy through emphasis on transparency and critique skills [8].

### **Successful Implementation Strategies**

Comprehensive training to develop faculty AI literacy knowledge addressing functionality, ethical, and societal concerns emerged as a key strategy [7]. K-12 contexts benefited from comprehensive frameworks including confidence and self-reflective mindsets [5]. Teacher education employed AI stoplight frameworks, professional learning institutes, faculty self-reflection surveys, institutional collaboration, and pedagogical innovation [1].

Computing education taught ethical use of generative AI and required explicit statements about its use in assessed work [4]. Programming assistants emphasized creating comprehensive tools, tracking student interactions, and pedagogical approaches [8].

Higher education developed structured roadmaps for AI integration, implementation toolkits, and emphasized institutional openness, capacity building, and acceptance of failure [2]. Translation programs used role-and-function-based models for custom GPTs with lightweight approaches for instructor workload [3].

Professional development integration into teacher education, continuous and contextualized training, and cultivation of innovation-supporting cultures represented successful approaches [10].

## **Impact on Teaching Practice**

### **Changes in Instructional Design and Delivery**

GenAI rendered traditional assignments and assessments obsolete, forcing fundamental instructional redesign [7]. K-12 frameworks proposed comprehensive approaches incorporating process and praxis design, focusing on student-centered learning and real-world applications [5]. Teacher education integrated AI into coursework using stoplight frameworks for transparency and reconfigured action research projects around AI capabilities [1].

Computing education experienced LLMs performing source code generation and interpretation, changing traditional teaching methods and assessment practices [4]. Educators across disciplines used AI to enhance instructional practices including differentiation, project-based learning, critical thinking, and explicit teaching [6].

Higher education saw AI transforming traditional teaching methods and assessments, making prior approaches obsolete [2]. Translation education fostered AI literacy and prompt engineering skills through custom GPTs enhancing strategic thinking [3]. Programming courses integrated AI tools to provide real-time support, preventing direct solutions [8].

AI offered new possibilities for personalized tutoring, automated feedback, and adaptive learning [9]. The requirement for teachers to rethink assessment and adopt new approaches to immersive learning represented fundamental pedagogical shifts [10].

### **Impact on Teacher Workload and Efficiency**

Faculty discomfort and neutrality toward GenAI indicated potential impacts on workload and efficiency as they struggled to adapt [7]. Teacher education efforts enhanced efficiency by streamlining administrative tasks [1]. Computing

education used LLMs to reduce instructor workload by generating personalized teaching materials and assignments [4].

AI supported instructionally grounded tasks, potentially reducing workload and increasing efficiency [6]. Teacher-focused tools reduced workload, allowing focus on strategic teaching aspects [2]. Translation education designed lightweight approaches for instructors, reducing workload and increasing time for coaching higher-order decisions [3]. Programming tools assisted with feedback and explanations, potentially reducing workload [8].

### **Effects on Teacher-Student Relationships and Classroom Dynamics**

Faculty's lack of comfort and understanding potentially affected teacher-student relationships and classroom dynamics [7]. Community engagement, hands-on activities, and presentations enhanced pedagogical capabilities and improved teacher-student relationships [5]. Teacher education created potential for more collaborative instructor-student time through personalized learning experiences [1].

Computing education used LLMs to assist students with assignments and provide detailed explanations, enhancing learning experiences [4]. AI acted as teaching assistant, expanding epistemic demand and enhancing pedagogical capabilities [6]. Teachers' roles shifted toward coaching or mentoring, promoting deeper engagement and diverse learning styles [2].

Translation education's teacher-machine-student triad recentered agency and accountability, promoting collaborative learning [3]. Programming education required customization by instructors to ensure AI tools supported learning without replacing human interaction [8]. AI reshaped instructional relationships and authority, affecting classroom engagement [9].

### **Enhanced Pedagogical Capabilities**

Teacher education demonstrated integration of AI into coursework and strategic AI leadership development [1]. Computing education enabled detailed code explanations and personalized teaching materials [4]. Teachers demonstrated emerging competencies in prompt design, instructional adaptation, and critical evaluation [6].

Higher education introduced "AI Wrangler" certification programs to equip educators with skills for ethical AI integration [2]. Translation education elevated instruction from rote memorization to dynamic proficiency through custom GPTs encouraging self-directed practice [3]. Programming education assisted in flipped classroom settings and self-reflection activities [8]. AI prompted educators to rethink teaching and feedback, moving toward hybrid intelligence models [9]. AI-powered platforms and virtual reality enhanced learning experiences [10].

### **Unintended Consequences and Challenges**

Challenges in adapting to GenAI required training to address faculty apprehensions [7]. Computing education faced concerns about students becoming over-reliant on LLMs, limiting learning and making educators' work more difficult [4]. Human oversight proved crucial to mitigate bias or lack of cultural relevance in AI outputs [6]. Teacher education surfaced concerns about data privacy, plagiarism, and unequal access to AI resources [1]. Programming education required educators to adapt teaching methods and ensure control over AI responses [8]. Tensions in teacher-student relations and ethical concerns required critical and reflexive engagement with AI [9].

## Student Learning Outcomes and Experiences

Evidence on student outcomes remained limited across studies, with most focusing on educator development rather than direct student impact. Where student outcomes were examined, several patterns emerged.

In K-12 AI education, frameworks aimed to improve literacy and competency by focusing on core knowledge, impact, ethics, collaboration, and self-reflection [5]. Five learning experiences—community engagement, global and local case studies, hands-on activities, exhibitions and presentations, and cultural learning—were designed to enhance student engagement and motivation [5]. The framework intended to develop student AI literacy and critical thinking skills through focus on core knowledge and self-reflection [5].

Teacher education contexts showed AI enhanced personalization in learning, offering tutoring support and specialized help for students with disabilities [1]. Students who used AI felt more confident in their abilities to conduct similar tasks in the future [1]. Students reported feeling more prepared to use Generative AI tools for teaching, with their work appearing more professional after using AI for source evaluation and literature review [1].

Computing education demonstrated that LLMs could solve introductory problems with ease, potentially rendering traditional teaching methods obsolete [4]. LLMs offered detailed code explanations and error message explanations, enhancing student understanding and engagement [4]. The tools facilitated renewed focus on problem-solving while providing personalized help [4].

Large-scale analysis of K-12 teacher-AI interactions suggested AI enhanced instructional practices, created or adapted content, supported assessment and feedback loops, and attended to student needs for tailored instruction, potentially improving academic achievement [6]. AI supported explicit teaching, inquiry-based learning, differentiation, and assessment, increasing student engagement and motivation [6]. AI encouraged critical thinking and high-level cognition through inquiry-based questions and context-responsive scaffolding, developing student critical thinking skills [6]. AI supported differentiated instructional strategies, potentially improving personalization and accessibility for diverse learners [6].

Higher education emphasized development of universal AI literacy including core competences such as prompting, bias detection, and ethics [2]. Frameworks recommended equity-centered implementation strategies like bias audits, device-lending programs, and multilingual AI options to ensure fairness and accessibility [2]. AI-enhanced pedagogy enhanced student engagement and motivation by combining human and AI strengths [2].

Translation education showed enhanced efficiency, rich learning resources, and improved critical reflection and revision skills [3]. Students experienced boosted motivation and confidence through personalized learning and immediate feedback [3]. The approach fostered AI literacy, prompt engineering, and ethical considerations [3]. It expanded accessibility to diverse learners, including those in under-resourced languages [3]. Students developed self-directed practice and collaborative skills [3].

Programming education with AI assistants deepened understanding, provided faster access to relevant knowledge, and offered new ways to learn code [8]. Students appreciated contextually relevant assistance and specific solutions [8]. The design promoted cognitive engagement and motivated learning by avoiding direct responses [8]. Women used CodeAid more often than men, indicating potential equity benefits [8].

AI offered personalized tutoring, automated feedback, and adaptive learning that reoriented pedagogy [9]. AI enhanced creative practice and critical thinking [9]. AI provided tailored educational experiences while affecting perceptions of authorship and integrity [9].

## **Ethical AI Frameworks and Implementation**

### **Specific Guidelines and Frameworks**

Emerging AI literacy definitions encompassed critical perspectives on bias, misinformation, energy consumption, academic integrity, data privacy, and intellectual property rights [7]. K-12 contexts employed IBM AI ethical principles focusing on fairness and biases, trust and transparency, accountability, social benefit, and privacy and security [5]. Teachers taught students to recognize biases and ensure fairness in AI applications, while encouraging examination of reliability and understanding privacy rights [5].

Teacher education referenced the Rome Call for AI Ethics emphasizing transparency, inclusion, reliability, impartiality, responsibility, and security [1]. Universities established policies and guidelines including data-protected access to Microsoft Copilot [1]. Computing education framed discussions around the ACM Code of Ethics, encouraging educators to teach ethical use and requiring students to include explicit statements about LLM use in assessed work [4].

Higher education AI competency frameworks included human-centered mindsets and ethics of AI [2]. Translation education emphasized structured integration of custom GPTs with human oversight to limit hallucinations and biases, aligning with human-centered AI paradigms [3]. Programming education implemented “guardrails” to prevent direct code solutions and used few-shot learning to control output [8]. Educators created structured spaces for ethical experimentation and integrated critical perspectives into teaching [9].

### **Data Privacy and Protection Measures**

Large-scale analysis platforms complied with FERPA, COPPA, GDPR, and SOC 2 standards, with IRB approval [6]. De-identification of data, removal of PII, and contractual terms preserved data privacy during analysis with commercial LLMs [6]. Higher education emphasized protecting student data privacy and mitigating socioeconomic disparities in AI access [2]. Translation education highlighted data privacy as paramount [3]. Programming education obtained student consent for data sharing and used anonymous session recordings [8].

### **Bias Detection and Mitigation**

K-12 education taught students to recognize biases and ensure fairness in AI applications [5]. Teacher education raised awareness of biases in algorithms and emphasized equitable access [1]. Higher education addressed bias in AI outputs and ensured transparency in AI use [2]. Translation programs recommended human oversight to mitigate biases [3]. Critical perspectives examined power, language, and representation, suggesting awareness of bias [9].

### **Transparency and Explainability**

K-12 contexts encouraged students to examine reliability of AI predictions and understand privacy rights [5]. Teacher education used stoplight frameworks and transparent instruction [1]. Computing education encouraged explicit statements about LLM use [4]. Higher education mandated AI tools in mental health support under clinician supervision with clear disclosures [2]. Translation education's teacher-machine-student triad recentered agency and accountability [3]. Programming education emphasized displaying incorrect responses and transparency in AI responses [8]. Creating structured spaces for ethical experimentation and integrating critical perspectives fostered transparency [9].



## Academic Integrity

Computing education held students responsible for LLM-generated content, requiring explicit statements about LLM use in assessed work, with users considered authors of generated text [4]. Higher education used AI detection tools for formative purposes only, with human review to prevent false positives [2]. Translation education adapted AI within pedagogical frameworks preparing students to be discerning and empowered [3]. Programming education focused on avoiding direct code solutions with emphasis on academic integrity in educator feedback [8]. Tensions around integrity and authorship required focused attention [9].

## Professional Responsibility and Accountability

Teacher education integrated AI critical engagement into coursework and provided leadership in faculty training [1]. Computing education required educators to teach ethical use, with students responsible for content [4]. Higher education emphasized human oversight and ethical safeguards [2]. Translation education positioned technology as augmenting human capabilities within HCAI paradigms [3]. Programming education reflected educators' concerns about AI tools and emphasized critique and transparency [8]. Fostering collaborative spirit to ensure AI serves inclusion, authenticity, and trust represented professional responsibility imperatives [9].

## Synthesis

The studies reveal substantial convergence around core principles of AI fluency for educators, despite heterogeneity in specific implementations. Three key insights emerge from reconciling apparent differences across contexts.

**Context-Specific Competency Emphasis:** The varying emphasis on technical versus ethical competencies reflects appropriate contextualization rather than conflicting frameworks. Computing education naturally prioritizes technical fluency with code generation and debugging [4, 8], while teacher education emphasizes pedagogical adaptation and equity [1]. K-12 contexts foreground student-centered confidence and self-reflection [5], whereas translation education focuses on domain-specific workflow integration [3]. These differences represent adaptive implementations of shared underlying principles—the 4D framework's elements manifest differently across disciplines but maintain consistent attention to delegation, prompting, discernment, and diligence.

**Progression Path Consistency Amid Implementation Diversity:** All frameworks converge on a three-tier progression (acquire/literacy → deepen/fluency → create/mastery), yet implementation approaches vary systematically by institutional capacity [1, 2, 4, 5]. Well-resourced institutions with robust professional development infrastructure implemented comprehensive certification programs and semester-long deployments [1, 2, 8], while others relied on iterative co-design with existing faculty [5]. The 75% faculty discomfort rate in one study [7] versus intensive 40+ training sessions in another [1] reflects not contradictory findings but differing institutional readiness levels. Both represent valid points on the same developmental continuum—institutions must match implementation intensity to current capacity while building toward comprehensive integration.

**Balancing Productivity Gains Against Ethical Safeguards:** The tension between AI's efficiency benefits and ethical risks resolves through implementation design rather than choosing one priority over another. Studies reporting productivity gains through automated grading and content generation [1, 2, 4, 6] simultaneously emphasized guardrails preventing over-reliance and maintaining academic integrity [3, 4, 8]. The solution lies in structured implementation: delegation of routine tasks paired with transparency requirements [1, 8], productivity tools coupled with human oversight [2, 3], and efficiency gains balanced by critical evaluation competencies [6, 8]. Studies showing successful integration consistently embedded ethical frameworks from inception [1, 2, 5] rather than treating ethics as an afterthought to productivity optimization.

The evidence suggests that effective AI fluency development requires: (1) context-appropriate frameworks that adapt shared competency models to disciplinary needs; (2) staged implementation matching institutional capacity while building toward comprehensive integration; and (3) dual-priority design embedding ethical safeguards within productivity-enhancing workflows rather than treating them as competing goals.

## References

1. Kelley MJ, Wenzel T (2025) Advancing Artificial Intelligence Literacy in Teacher Education Through Professional Partnership Inquiry. *Education sciences*. <https://doi.org/10.3390/educsci15060659>
2. Petrova MN (2026) Strategies for developing AI competencies in higher education. *Frontiers in Education*. <https://doi.org/10.3389/feduc.2025.1683909>
3. Karaban V, Karaban A (2025) AI Era Classroom Beyond Post-Editing: Custom Gpts in Translator Training. *The Journal of VN Karazin Kharkiv National University Series: Foreign Philology Methods of Foreign Language Teaching*. <https://doi.org/10.26565/2786-5312-2025-102-11>
4. Prather J, Denny P, Leinonen J, et al (2023) The Robots Are Here: Navigating the Generative AI Revolution in Computing Education. *ITiCSE-WGR*. <https://doi.org/10.1145/3623762.3633499>
5. Chiu T, Ahmad Z, Ismailov M, Sanusi I (2024) What are artificial intelligence literacy and competency? A comprehensive framework to support them. *Computers and Education Open*. <https://doi.org/10.1016/j.caeo.2024.100171>
6. Liu AX, Esbenshade L, Sarkar S, et al (2025) How K-12 Educators Use AI: LLM-Assisted Qualitative Analysis at Scale. *ArXiv*
7. Neumann M, Gerstl-Pepin C (2025) Faculty Responses to Generative AI: The Shifting Landscape of Higher Education. *International Symposium on Technology and Society*. <https://doi.org/10.1109/ISTAS65609.2025.11269643>
8. Kazemitabaar M, Ye R, Wang X, et al (2024) CodeAid: Evaluating a Classroom Deployment of an LLM-based Programming Assistant that Balances Student and Educator Needs. *International Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3613904.3642773>
9. Zabrodskaia A, Girardelli D, D MJG, et al (2026) Editorial: Teaching and assessing with AI: teaching ideas, research, and reflections. *Frontiers in Communication*. <https://doi.org/10.3389/fcomm.2025.1769019>
10. Ngongpah G, Oni O (2025) Teachers' Readiness and Competency in Using AI in the Classroom. *Asian Journal of Education and Social Studies*. <https://doi.org/10.9734/ajess/2025/v51i92412>